Ruminant Disease

Digestive Physiology of the Ruminant



Herbivores

Herbivores are also known as plant-eating animals, which can be further classified into graminivores (食草動物), folivores (食葉動物), frugivores (食果動物), and granivores (食穀動物), etc. In fact, many herbivores consume various parts of a plant, such as the roots and seeds. The dietary habits of some herbivores change with the seasons, especially in temperate regions, where different foods are available at different times of the year.

Taxonomic Rank

Cattle

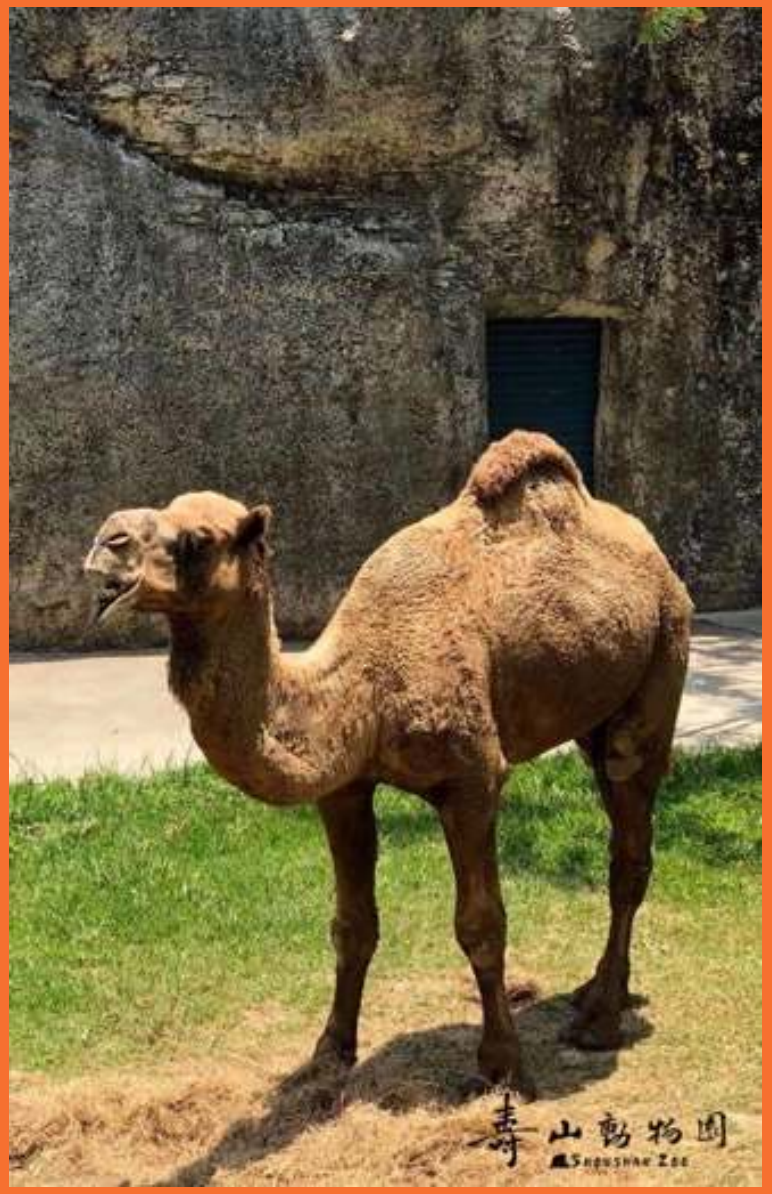
- Kingdom: Animalia
- Phylum: Chordata
- Class: Mammalia
- Order: Artiodactyla
  - Suborder: Ruminantia
- Family: Bovidae
  - Subfamily: Bovinae

Taxonomic Rank

Cattle

- kingdom: Animalia
- phylum: Chordata
- class: Mammalia
- order: Artiodactyla
  - sub-order: Ruminantia (反芻亞目)
    Suina (豬型亞目)
    Tylopoda (胼足亞目)

– kingdom: Animalia
– phylum: Chordata
– class: Mammalia
– order: Artiodactyla
  – sub-order: Tylopoda
– family: Camelidae
– genus: *Camelus* (駱駝屬)
  Lama (羊駝屬)







Taxonomic Rank

Cattle

– order: Artiodactyla

  – sub-order: Ruminantia

– family: Bovidae (牛科/洞角科)

  – sub-family: Bovinae

– genus: *Bos* (牛屬)

– species: *Bos taurus* (家牛)

  – sub-species: *B. t. primigenius* (原牛)

    B. t. taurus (歐洲牛)

    B. t. indicus (瘤牛)

Bos taurus ssp. *taurus*



https://cattleinternationalseries.weebly.com/holstein.html

Bos taurus ssp. *indicus*



Taxonomic Rank

- family: Bovidae (牛科/洞角科)
  - sub-family: Bovinae
- genus: *Bubalus* (水牛屬)
- species: *Bubalus bubalis* (亞洲水牛)
  - sub-species:
    - *B. b. arnee* (河川水牛)
    - *B. b. carabanesis* (沼澤水牛)



https://a-z-animals.com/animals/water-buffalo/



By Kleomarlo

Water buffalo

Bubalus bubalis can be further divided into swamp and river types. The swamp buffalo (*B. b. carabanesis*) has 48 chromosomes, is primarily distributed in Southeast Asia, and is mostly used as a draft animal. The river buffalo (*B. b. arnee*) has 50 chromosomes, is primarily distributed in India, Italy, and the Middle East, and is mainly used for dairy production.

Taxonomic Rank

Cattle

– order: Artiodactyla

  – sub-order: Ruminantia

– family: Bovidae (牛科/洞角科)

  – sub-family: Bovinae

    Caprinae (羊亞科)

– genus: *Ovis* (羊屬)

  Capra (山羊屬)

Ruminants

Ruminants (order Artiodactyla, sub-order Ruminantia) are even-toed, herbivorous grazing or browsing mammals. They have the ability to regurgitate ingested food back into their mouths and chew it again before sending it to a specialized stomach (such as the rumen). By microbial fermentation, they are able to extract nutrients from plant-based foods. Because this process occurs in the front part of the digestive system, it is known as foregut fermentation.

Types of Digestive Systems

Monogastrics


Dogs


Pigs


Cats


Horses


Human

Ruminants


Beef Cattle


Dairy Cattle


Goats


Sheep


Deer

Poultry


Chickens


Turkeys


Ostrich

http://images.slideplayer.com/15/4685806/slides/slide_4.jpg



Contents

1. Dairy cattle production lifecycle
2. Nutrition and feeding management
3. Transition period
4. Dietary cation-anion difference (DCAD) & hypocalcemia

Dairy Production Medicine

With the growing global focus on the food safety of animal dairy and meat products, the critical issue of antimicrobial resistance, and the emergence of novel and zoonotic diseases, the role of modern farm veterinarians has shifted from focusing solely on the diagnosis and treatment of individual animal diseases.





Dairy Production Medicine

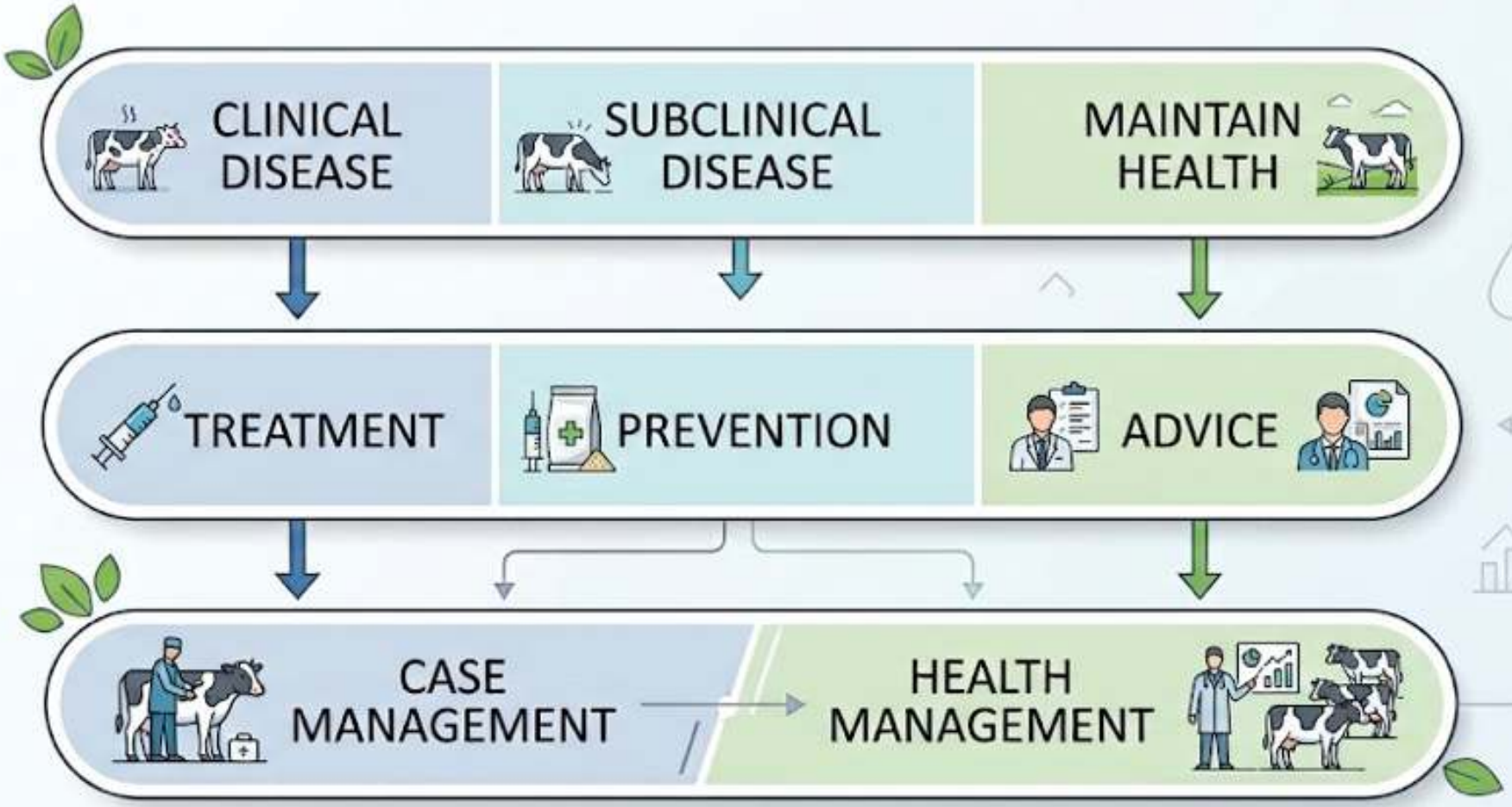
The scope of work for large animal veterinarians must evolve beyond traditional disease diagnosis skills into an integration of knowledge encompassing veterinary epidemiology, livestock facility engineering, animal nutrition, farm management economics, personnel management, and record-keeping.

Dairy Production Medicine

Furthermore, we must understand the entire “farm-to-table” continuum to assist producers in achieving their annual farm management goals. This represents the new concept known as production medicine.



DEVELOPMENT OF THE DAIRY VETERINARY MEDICINE



MODERN DAIRY VETERINARY CONCEPT

Livestock Veterinarians vs. Production Medicine

Among almost all livestock veterinarians, the clinical work of those specializing in cattle and sheep/goat is unique. Beyond diagnosing and treating diseases in **individual animals**, they must also specifically focus on the management and monitoring of **herd health**. By preventing the occurrence of diseases, they can effectively enhance the economic profitability of the farm.

Dairy Production Medicine

Dairy farming is an intensive-livestock industry aimed at rearing cows with high milk yields and good milk quality to achieve optimal economic profitability. However, accomplishing this goal relies on:

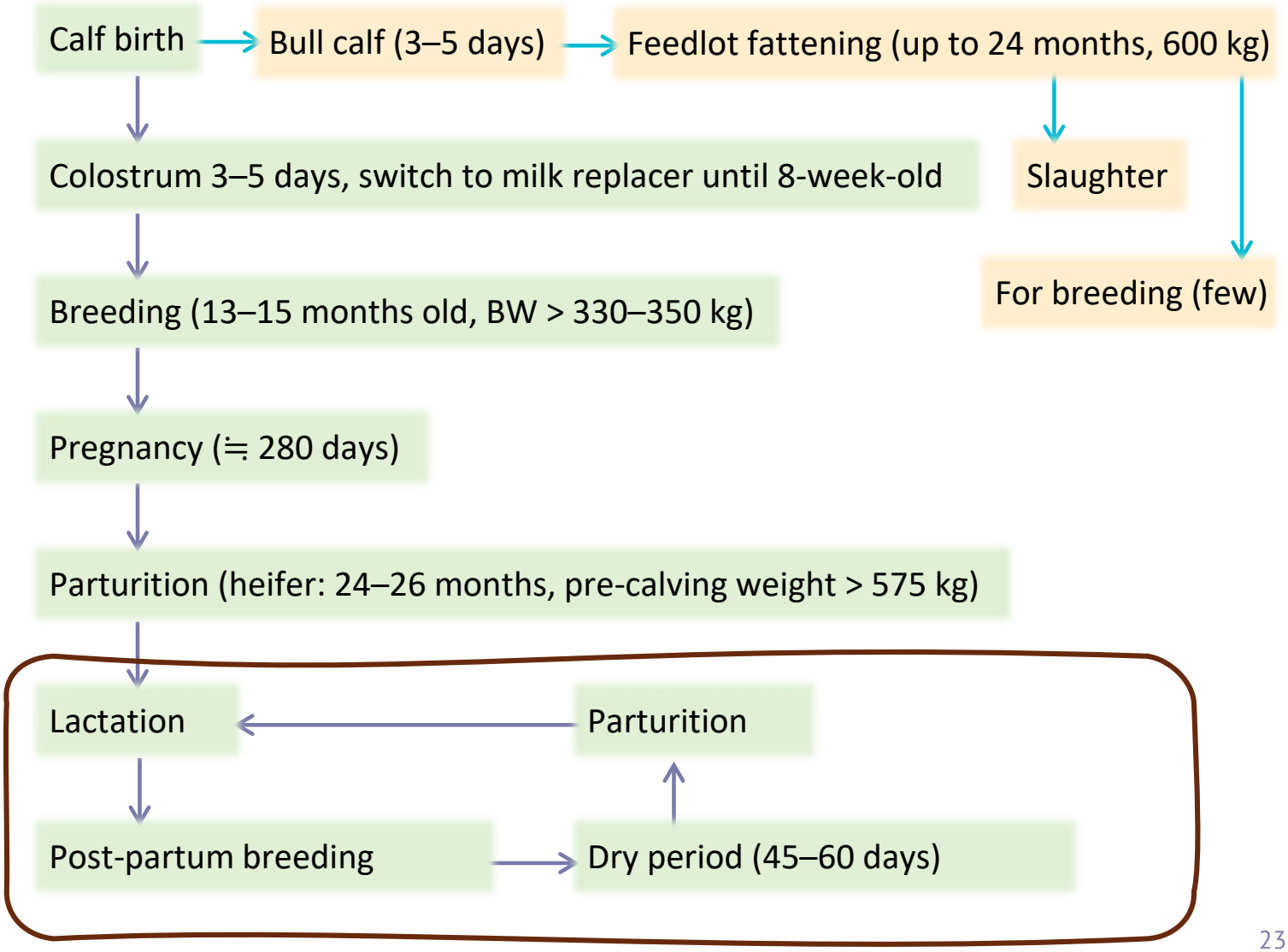
Dairy Production Medicine

- **Good feeding and management concepts and techniques**, which include balanced nutritional rations, proper management, and disease prevention and control;
- **A suitable rearing environment**, encompassing dairy housing design and other related facilities;
- **Genetics and breeding**.

Dairy Production Medicine

Dairy farming is an intensive-livestock industry aimed at rearing cows with high milk yields and good milk quality to achieve optimal economic profitability. However, accomplishing this goal relies on:

Dairy cattle production lifecycle




Nutrition &
feeding management

Key Indicators for Dairy Management

- Maintain maximum dry matter intake (DMI)
- Balanced ration formation for dairy animals
- Maximum digestibility and absorption rates

Essential Nutrients

- Water: the most important
- Energy: carbohydrates and lipids
- Protein: rumen degradable and rumen undegradable protein (RDP & RUP), amino acids
- Vitamins: A, D, and E
- Minerals: macro, micro, DCAD

Essential Nutrients—Water

A lactating dairy cow requires 4–5 L of water per kg of DMI, and an additional 3–5 L of water for every kg of milk produced. A non-lactating cow (with a DMI of 10 kg/day) drinks approximately 40 L of water daily. However, when producing 16–20 kg of milk per day, it requires an additional 70 L of water; for a cow yielding 35 kg/d, total water consumption can approach 100 L.

Daily water intake (L) = Dry matter intake per day in kg x 6 + milk yield

Essential Nutrients—Water

Environmental temperature also has a significant impact on the water intake of dairy cows. For instance, in high-temperature environments or when cows are experiencing heat stress, water intake can increase to 10–12 L per kg of DMI. Conversely, in cooler environments (< 20°C), the water requirement drops to approximately 3 to 4 L per kg of DMI.

Essential Nutrients—Water

High-quality drinking water must be fresh, clean, and cool. Furthermore, it must be easily accessible to the cows, allowing them to drink *ad libitum* (freely at any time).





Essential Nutrients—Energy

Energy is the driving force behind all body functions, and carbohydrates in the feed serve as the primary energy source for dairy cows. Additionally, fats (or lipids) can provide a portion of the required energy, though they are not strictly essential. The energy requirements of dairy cows are commonly expressed as Net Energy for Lactation (NEL).

Producing 1 kg of raw milk (4% fat) requires 0.74 Mcal of NEL.

Nutrition and feeding of dairy cattle

Peter S. Erickson[a], *Kenneth F. Kalscheur*[b]

[a]Department of Agriculture, Nutrition, and Food Systems, University of New Hampshire, Durham, NH, United States; [b]U.S. Dairy Forage Research Center, USDA-ARS, Madison, WI, United States

TABLE 9.4 Nutrient requirements of lactating dairy cattle.[19]

(NRC, 2001)

Milk production, kg	**25**		**54.4**	
DM intake[a], kg/day	20.3		30.0	
Energy, NE_l[b], Mcal/kg	1.37	(27.8)	1.61	(48.3)
Metabolizable protein, g/d	1,862		3,476	
RDP[c], %	9.5		9.8	
RUP[d], %	4.6		6.9	
NDF[e], % min	25–33		25–33	
NFC[f], % max	36–44		36–44	

[a] *DM intake = dry matter intake.*
[b] *NE_l = net energy of lactation.*
[c] *RDP = rumen degradable protein.*
[d] *RUP = rumen undegradable protein.*
[e] *NDF = neutral detergent fiber.*
[f] *NFC = non-fiber carbohydrate.*

https://pmc.ncbi.nlm.nih.gov/articles/PMC7153313/pdf/main.pdf

Feeds for productive animals

Feeds are divided into two main categories: concentrates and roughages (or forages). Feeds that are high in fiber and low in energy are classified as forages. Concentrates, on the other hand, are higher in carbohydrates or proteins and lower in fiber. Examples of concentrates include corn, barley, and soybean meal.





Forage

The main forages use in Taiwan can be broadly classified into two types: grasses (禾本科) and legumes (豆科).

Examples of grasses include Bermuda hay (百慕達草), oat hay (燕麥), Pangola hay (盤固拉草), and Timothy hay (梯牧), as well as green-chop corn (青割玉米) and Napier grass (狼尾草). Legumes are represented by forages such as alfalfa hay (苜蓿草).

Forage

In general, legumes have a higher protein and mineral content than grasses. The rhizobia (root nodule bacteria, 固氮根瘤菌) associated with legumes are capable of utilizing atmospheric nitrogen to synthesize plant proteins. Grasses lack these bacteria, which accounts for their lower protein content.

Forages

狼尾草Napier Grass



(From PlantVillage)



https://milnepublishing.geneseo.edu/botany/chapter/corn/

玉米
Corn

Forages

苜蓿Alfalfa



百慕達Bermuda hay



燕麥草Oat hay



盤固拉Pangola hay



梯牧草Timothy hay



牧草行情參考表

(2026. 08)

牧草種類	草商A參考價格(元/kg)	草商B參考價格(元/kg)	酪農詢價
苜　蓿	17.0～20.0	14.7～17.5	15.2～18.5
甜燕麥	14.0～17.0	14.1～15.2	14.5～16.8
百慕達	13.0～15.0	12.6～14.9	12.8～14.8
國產牧草種類	草農a參考價格(元/kg)	草農b參考價格(元/kg)	酪農詢價
青刈玉米	缺貨中（非產期）		
青貯玉米	3.5～3.6	3.4～3.5	3.4～3.6
狼尾草	1.7～1.8	1.7～1.8	1.7～1.8

因各地交易條件及運費等因素之不同，故以上價格僅供參考。

Total mixed ration (TMR)



TMR is a complete diet formulated based on the nutrient requirements. It blends all feed ingredients—including concentrates, high-quality forages, and agricultural by-products—into a uniform mixture fed *ad libitum*. This system differs from traditional feeding methods, which offer concentrates and forages separately.

表、牧場TMR電腦配方實例(泌乳牛共106頭)

	低產	53頭	高產	53頭
	日配量(kg)	平均(kg)	日配量(kg)	平均(kg)
百慕達乾草	110	2.1	125	2.4
梯牧乾草	50	0.9	50	0.9
苜蓿乾草	185	3.5	200	3.8
燕麥乾草	200	3.8	200	3.8
飼料	180	3.4	190	3.6
玉米酒粕	50	0.9	50	0.9
豆殼	50	0.9	50	0.9
麩皮	30	0.6	30	0.6
玉米片	100	1.9	140	2.6
啤酒粕	200	3.8	200	3.8





Benefits of Implementing TMR Feeding System

1. Uniform and balanced nutrient intake:
   As a complete diet, it ensures cows receive a uniform and nutritionally balanced feed intake.
2. Prevents sorting and selective eating:
   It prevents cows from sorting through the feed and picking out specific ingredients.
3. Consistent rumen fermentation:
   Balanced intake stabilizes rumen fermentation, which increases feed efficiency, boosts milk yield and daily weight gain, and improves milk and meat quality.

Benefits of Implementing TMR Feeding System

4. Improves feed palatability:
   It enhances the overall taste and acceptance of the diet.
5. Reduces labor and improves operational efficiency:
   It saves labor costs and enhances operational efficiency, especially when integrated with mechanization.

Essential Nutrients—Energy

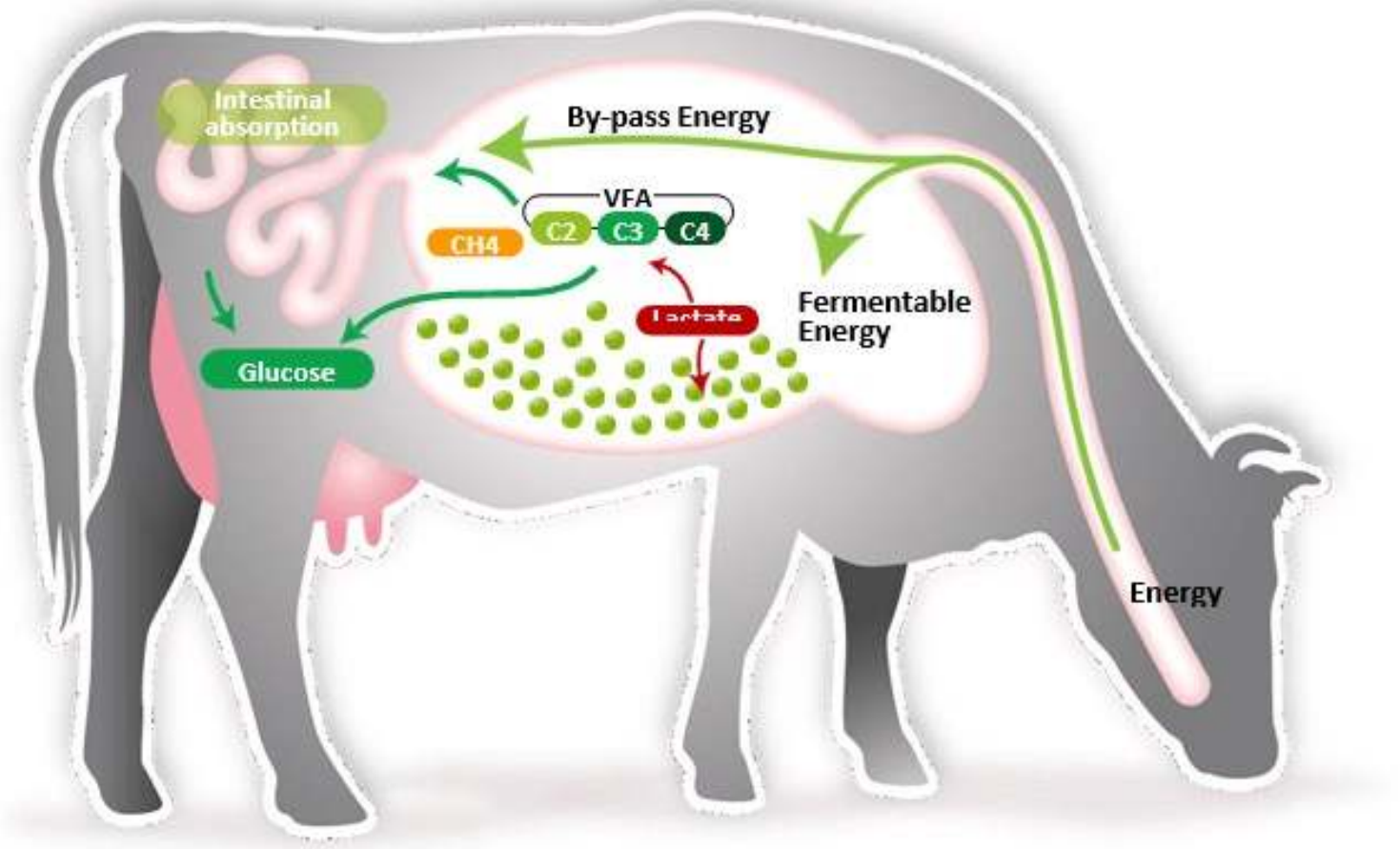
Dietary carbohydrates are fermented by rumen microbes to produce short-chain fatty acids (SCFAs) or volatile fatty acids (VFAs), which serve as the primary energy source and as substrates for glucose synthesis (gluconeogenesis) in dairy cows.

Essential Nutrients—Energy

These VFAs mainly consist of acetate (≒ 65%), propionate (≒ 15%), and butyrate (≒ 10%). The VFAs produced in the rumen account for about 84% of the total VFAs generated throughout the entire digestive tract. In monogastric animals such as pigs, carbohydrates are absorbed and utilized as monosaccharides. Therefore, monogastric animals can be referred to as “sugar animals”, whereas ruminants are considered “fatty acid animals”.

Intestinal digestibility is also key for an optimal energy and diet efficiency

In the rumen, energy of the diet is transformed in VFA (glucose precursor)



https://www.srpublication.com/energy-efficiency-in-ruminant/

Essential Nutrients—Energy

Carbohydrates utilizable by dairy cows can be broadly classified into two categories. One consists of structural carbohydrates derived from forage fiber. They have a dense structure and are fermented by rumen microbes at a slower rate, yielding a lower NEL. However, they produce a higher proportion of acetate among the VFAs. Since acetate serves as the primary precursor for milk fat synthesis, these carbohydrates contribute to increasing the milk fat percentage.

Essential Nutrients—Energy

Carbohydrates utilizable by dairy cows can be broadly classified into two categories. One consists of structural carbohydrates derived from forage fiber. They have a dense structure and are fermented by rumen microbes at a slower rate, yielding a lower NEL. However, they produce a higher proportion of acetate. Since acetate serves as the primary precursor for milk fat synthesis, these carbohydrates contribute to increasing the milk fat percentage.



Essential Nutrients—Protein

The dietary protein consumed by dairy cows can be divided into three fractions:

- Rumen degradable protein (RDP) → Protein / Non-protein nitrogen (NPN)
- Rumen undegradable protein (RUP)
- Indigestible intake protein (IIP)

Rumen degradable protein (RDP)

Rumen microbes play a critical role in the digestion and utilization of dietary protein. After the ingestion of dietary crude protein (CP, a collective term for nitrogenous substances), approximately 60–65% or more is degraded by microbes into ammonia. This ammonia serves as the required nitrogen source for microbial proliferation and is subsequently synthesized into bacterial protein or amino acids.

Rumen degradable protein (RDP)

This bacterial protein passes into the small intestine, where it is digested and utilized by the cow. According to the recommendations by NASEM (2021) (formerly NRC, the National Academies of Sciences, Engineering, and Medicine), for a dairy cow yielding 30 kg/d, RDP should account for about 60–65% of the dietary CP, while RUP should account for approximately 35–40% of the CP.

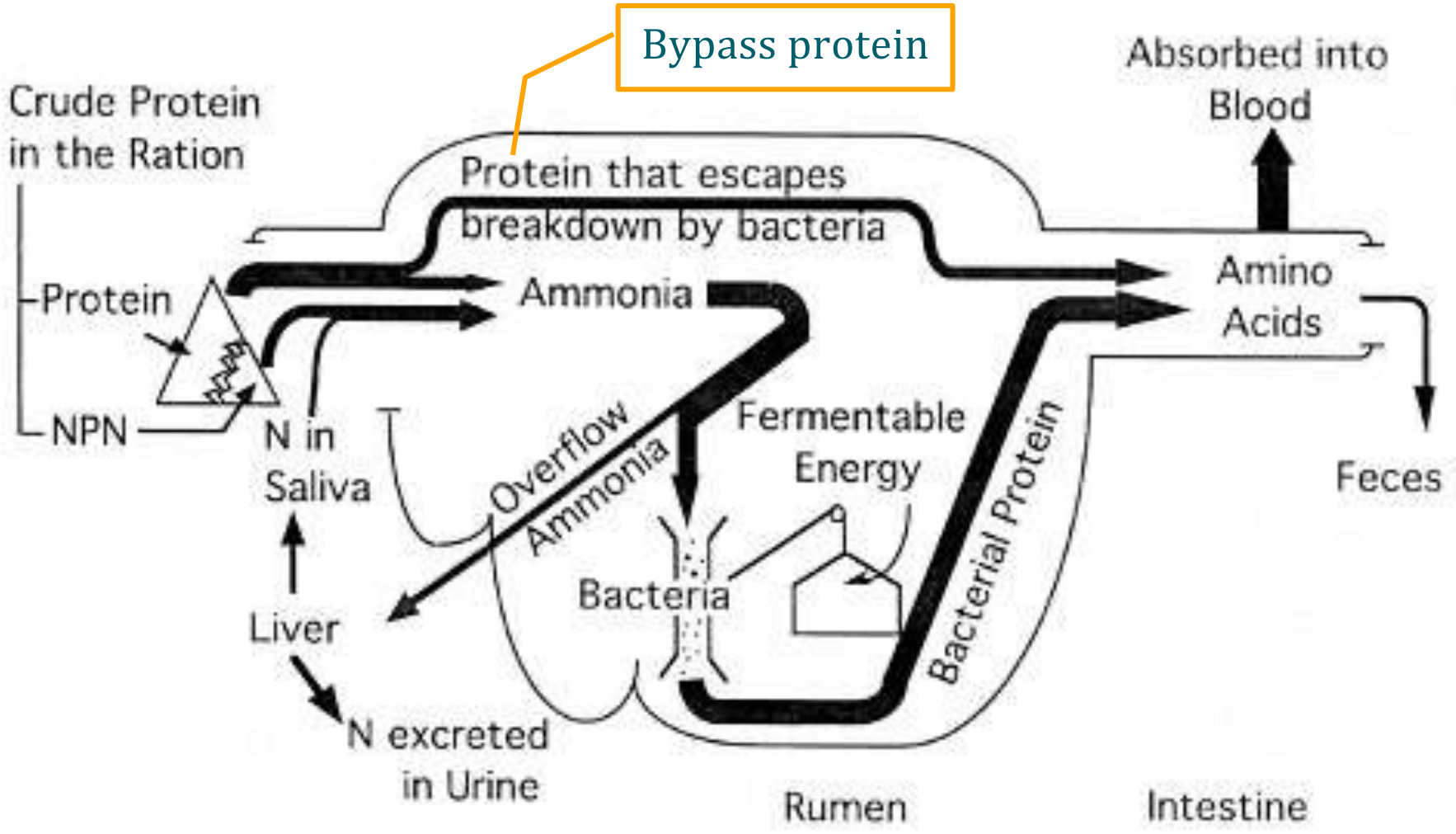
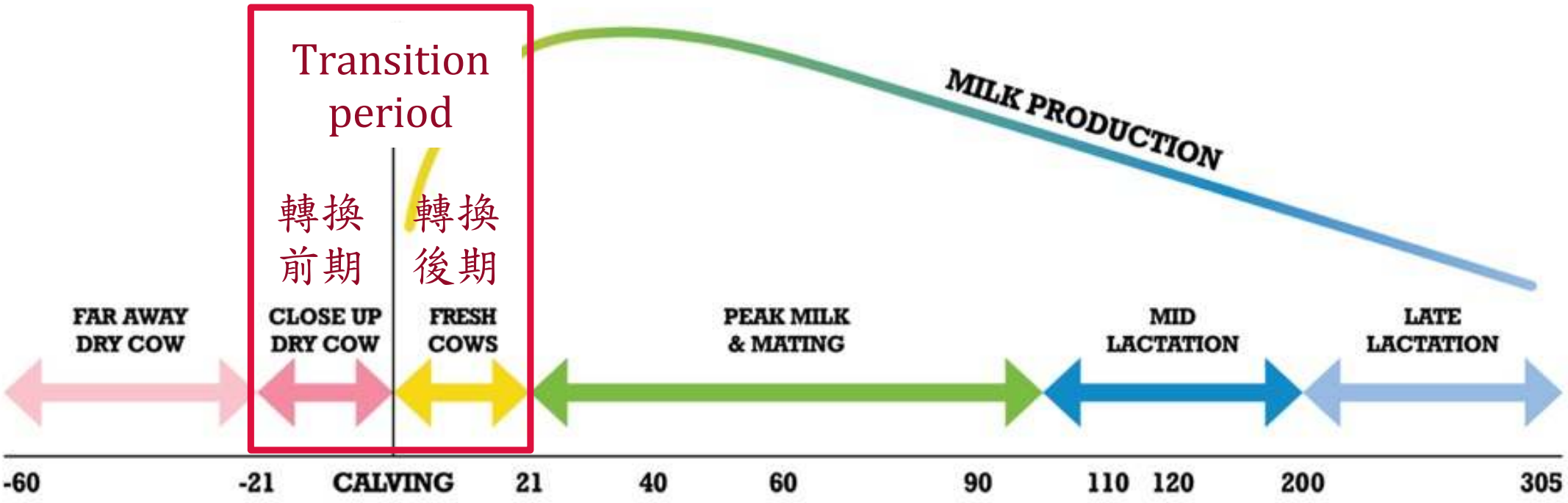


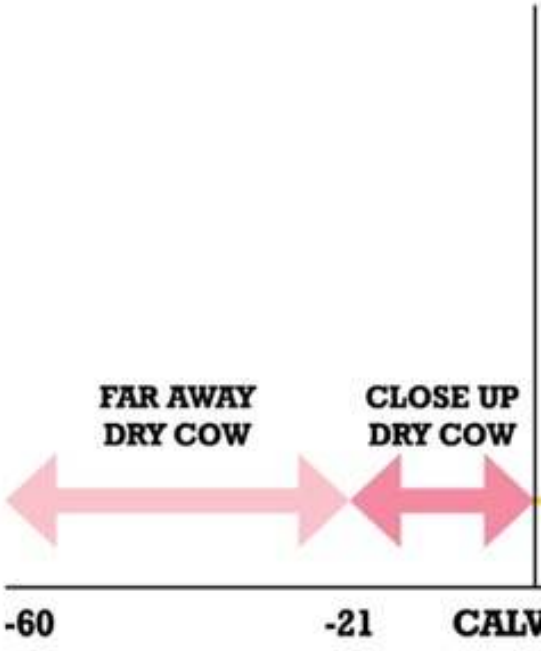
Figure. Schematic summary of nitrogen utilization by the dairy cow and other ruminants.
(Source: Satter and Roffler, 1975)

Phases of Lactation



https://www.healthycow.co.nz/importance-of-calcium-for-cows

Dry Period



cows are dried off, a stage that refers to the cessation of lactation. The ideal dry period lasts 45–60 days and encompasses three critical objectives: (1) mammary gland repair and regeneration, (2) meeting the demands of rapid fetal development, and (3) colostrum synthesis. Additionally, this period allows cows to manage their body condition, accumulating body fat to cope with the postpartum negative energy balance.

Transition Period

From 21 days pre-calving to 21 days post-calving, a cow's physiology transitions from the dry period to lactation. However, farmers frequently overlook the importance of feeding and managing of this specific group, viewing them as non-lactating cows that do not yet generate economic returns.

Principles for Dry Cow Management

- early dry period (far-off): high-fiber and low-energy diets
- late dry period (close-up): higher energy and lower fiber levels

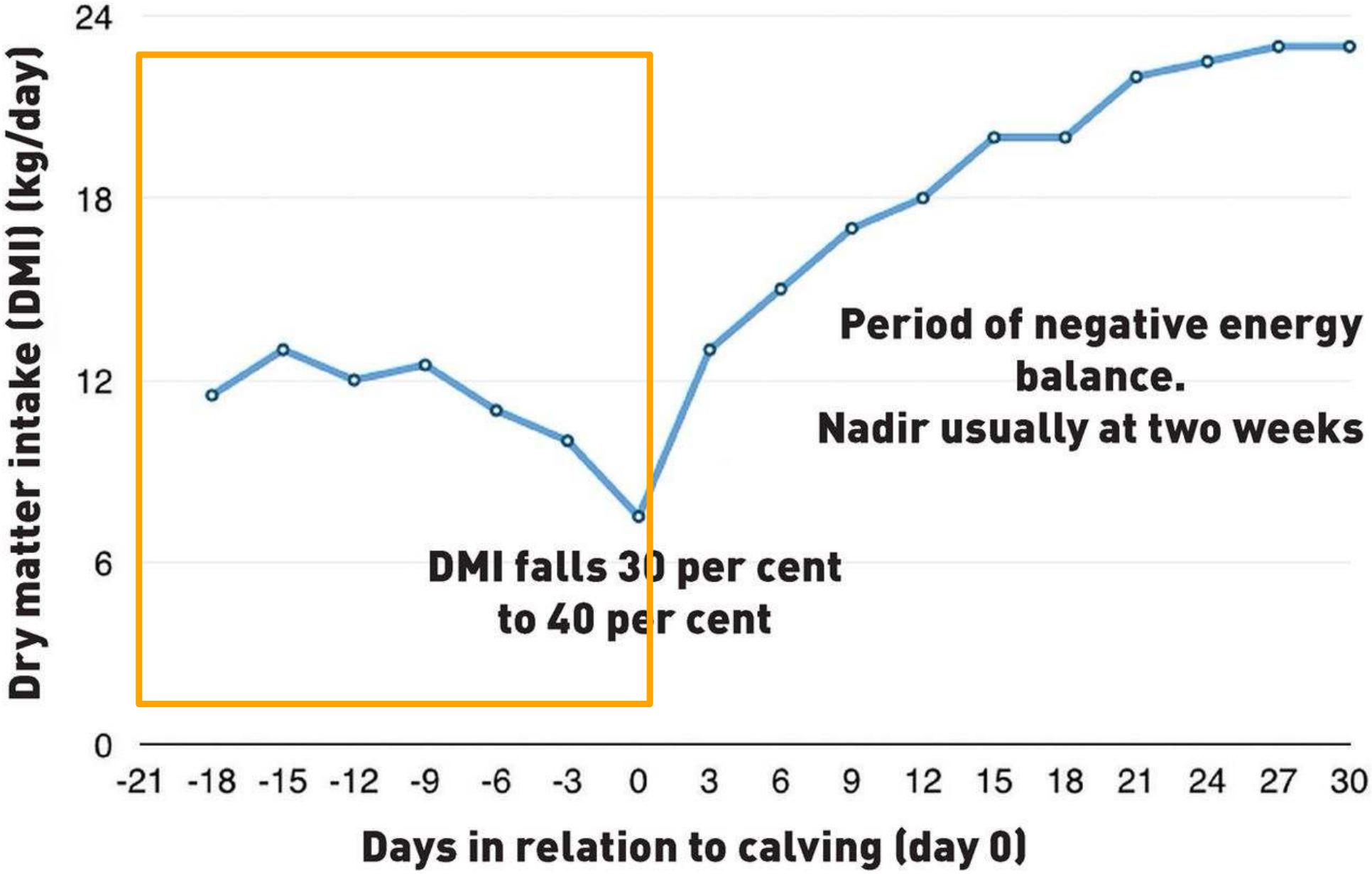
Providing 4–5 kg of concentrates (15–16% CP) in the late dry stage helps the rumen papillae and microbiome adapt to the impending high-concentrate lactation diet.

NUTRIENT GUIDELINES FOR TRANSITION RATIONS			
	Far-off cow	Close-up dry cow	Fresh cow
Ne_l (Mcal/lb)	.60	.68 - .70	.77
Crude protein, %	12	15 - 16	18
NDF, %	45 - 65	35 - 40	30
Nonfiber carbohydrate, %	25	30	35
Fat, %	3	4	5

Close-up dry period



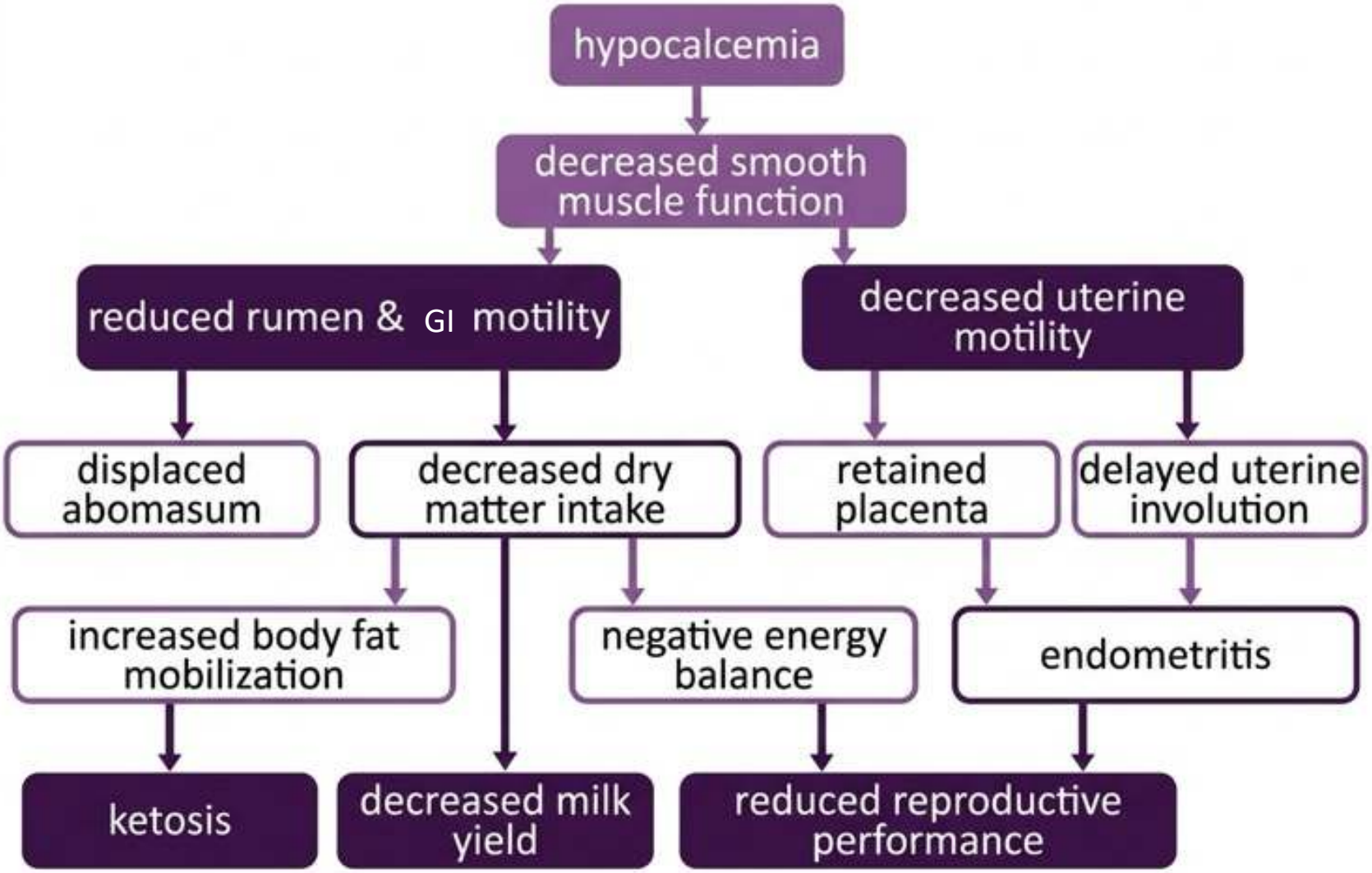
In the last three weeks before calving, the cow's immune function begins to decline. It is crucial to maximize DMI during this period because fetal nutritional demands are peaking, yet the cow's own DMI decreases by 30–40%.


Dry matter intake (DMI) (kg/day)
24
18
12
6
0
-21 -18 -15 -12 -9 -6 -3 0 3 6 9 12 15 18 21 24 27 30
Days in relation to calving (day 0)
DMI falls 30 per cent
to 40 per cent
Period of negative energy
balance.
Nadir usually at two weeks

Hypocalcemia / milk fever

Hypocalcemia, or milk fever, is a common metabolic disorder in dairy cows characterized by low blood calcium levels. It presents in both clinical and subclinical forms, severely impacting cow health during their most vulnerable stage: the transition period. Immediately after calving, dairy cows require a massive amount of Ca. They initially draw this Ca from the bloodstream, and subsequently mobilize it from their diet and bone reserves. If unable to mobilize sufficient Ca, affected cows may become unable to move or stand normally and will experience a decreased appetite, ultimately leading to compromised overall health and poor lactation performance.

Postpartum hypocalcemia syndrome in dairy cows



Dietary cation-anion difference (DCAD)

DCAD is a specialized dietary metric for ruminants, typically expressed in milliequivalents (mEq).
The calculation formula is: $(Na^+ + K^+) - (Cl^- + S^-)$.

Feeding a negative DCAD diet mildly acidifies the blood, stimulating the cow to mobilize stored Ca from her bones. This acts as an internal regulatory mechanism for Ca homeostasis, preparing the cow to cope with the massive loss of Ca associated with lactation.


Thank you.